Supplementary Information

Geographic and Linguistic Bias in LLMs for Air Pollution Policy:
A 25-Country, 14-Model, 4-Language, 5-Task Benchmark

Reproducibility. Every data-driven table below (Tables S1–S7) is regenerated from the raw confirmatory artifacts by `code/analysis/make_supplement_tables.py`; no value is hand-entered. Scoring is adjudicated by code against the official register wherever the correct answer is a checkable value (T2, T3), and by the mean of a three-vendor judge panel (Gemini 2.5 Pro, Claude Sonnet 4.6, DeepSeek-V3) elsewhere; T1 and T5 retain the scores of the original single judge (`gpt-5-mini-2025-08-07`). The analysed confirmatory corpus comprises 9,251 scored responses (7,580 English-prompt; 1,671 native-language) over 25 countries and 14 deployment stacks. All inputs, the scoring code, the ground-truth registry, and the anonymised responses will be released on publication under CC-BY-4.0 (data/prompts) and MIT (code).

Contents

S1 Pre-specification, design, and country selection

The hypotheses, primary models, persona protocol, exclusion criteria, multiple-comparison corrections, and the country sampling frame were specified in advance of confirmatory data collection. The pre-specified design covers 15 countries; the sample was subsequently extended, *post registration*, to 25 countries to gain power for the development gradient (H1) and the corpus-mechanism tests (H4) and to multiply the within-country language pairs (H2). Throughout the manuscript and this supplement, every effect computed on the 25-country sample is reported alongside its pre-specified 15-country value, and no pre-specified conclusion is revised on the basis of the extension alone (Section S8).

Countries were selected by theoretical sampling triangulated across three independent classifications: the UNCTAD Global North/South classification (operationalising the coloniality-of-knowledge framing), the Joshi et al. (2020) six-class linguistic-resource taxonomy (Joshi et al., 2020), and World Bank income groups. The pre-specified 15 (Set "P" in Table S4) span four world regions, five Joshi classes, and four income groups; the 10 extension countries (Set "E") comprise seven additional Global North references (United Kingdom, Canada, Australia, South Korea, France, Italy, Portugal) chosen to thicken the high-development end of the gradient, and three additional Global South native-language-pair countries (Colombia, Chile, Angola). The full design rationale is in `docs/supplement/s1_design_rationale.md`.

S2 Deployment-stack roster and serving configuration

Unit of analysis. We evaluate each model *as served* through a fixed access stack — the tuple of model weights, vendor, serving infrastructure, and API/endpoint layer — rather than idealized weights in isolation, because a public manager interacts with the served stack and any measured gap is a property of that stack. Table S1 lists the 14 stacks with their exact API/tag strings and execution venues.

Table S1. Deployment-stack roster. Each model is evaluated *as served* through a fixed access stack; the API/tag string and execution venue are part of what each label denotes. Parameter counts are vendor-reported (dense or total for MoE; closed-frontier counts undisclosed). N and mean are realized English-prompt confirmatory coverage.

Model	Vendor	Tier	API / tag string	Venue	N	Mean
DeepSeek-V3	DeepSeek	A	`deepseek-chat`	DeepSeek API	500	0.625
Command R+	Cohere	A	`command-r-plus-08-2024`	Cohere (trial)	500	0.509
Llama 3.3 70B	Meta	A	`llama-3.3-70b-versatile`	Groq (free)	474	0.506
GPT-OSS 120B	OpenAI	A	`openai/gpt-oss-120b`	Groq (free)	298	0.485
Llama 4 Scout	Meta	A	`meta-llama/llama-4-scout-17b-16e-instruct`	Groq (free)	500	0.473
Qwen3 32B	Alibaba	B	`qwen/qwen3-32b`	Groq (free)	498	0.546
Phi-4 14B	Microsoft	B	`phi4`	Ollama (local)	500	0.485
Qwen3 14B	Alibaba	B	`qwen3:14b`	Ollama (local)	500	0.474
Llama 3.1 8B	Meta	C	`llama-3.1-8b-instant`	Groq (free)	777	0.424

Model	Vendor	Tier	API / tag string	Venue	N	Mean
Cabra-Mistral 7B	botbot-ai	C	`cabra-mistral-7b`	Ollama (local)	496	0.320
GPT-5-mini	OpenAI	D	`gpt-5-mini`	OpenAI API	777	0.626
Gemini 2.5 Flash	Google	D	`gemini-2.5-flash`	Google AI (free)	490	0.604
Claude Haiku 4.5	Anthropic	D	`claude-haiku-4-5`	Anthropic API	490	0.597
GPT-5	OpenAI	E	`gpt-5`	OpenAI API	780	0.647

Determinism and seed handling. All stacks were queried at temperature 0.3 with the maximum-determinism configuration each provider documents, not an identical parameter surface, because the surfaces differ. Seed support is heterogeneous: locally served stacks (Ollama) honour a fixed seed; OpenAI exposes an advisory seed; Anthropic's Messages API exposes no seed parameter; and for the remaining hosted endpoints a seed is either advisory or not exposed. For every call we record the requested seed and a `seed_status` field (`sent`, `logged-only`, or `not-supported`) in the released run metadata, so that the determinism actually achieved is auditable rather than assumed. Two stochastic replications per (prompt, model, persona) cell estimate within-cell variability.

Acquisition window. Served-model behaviour can drift as vendors update infrastructure, so the reported accuracies are point-in-time estimates relative to the collection window. The pinned tag, version string, and acquisition timestamp are recorded for every call. Confirmatory collection ran from 2026-04-26 to 2026-06-12 (UTC), spanning the pilot, the pre-specified 15-country collection, and the post-hoc extension to 25 countries. Table S2 gives the first and last non-error call timestamp per provider together with the official API documentation consulted; the full per-call timestamps are released with the dataset.

Table S2. Per-provider acquisition window and official API documentation. Reported accuracies are point-in-time estimates relative to this window; first/last are the earliest and latest non-error call timestamps (UTC) recorded in the run logs. The window spans the pilot, the pre-registered 15-country collection, and the post-registration extension.

Provider (venue)	First (UTC)	Last (UTC)	API documentation
Anthropic API	2026-04-27	2026-06-12	`https://docs.anthropic.com/en/api/messages`
Cohere (trial)	2026-04-26	2026-06-11	`https://docs.cohere.com/reference/chat`
DeepSeek API	2026-04-26	2026-06-11	`https://api-docs.deepseek.com`
Google AI (free)	2026-04-27	2026-06-11	`https://ai.google.dev/api`

Provider (venue)	First (UTC)	Last (UTC)	API documentation
Groq (free)	2026-04-26	2026-06-11	https://console.groq.com/docs/api-reference
Ollama (local)	2026-04-27	2026-06-12	https://github.com/ollama/ollama/blob/main/docs/api.md
OpenAI API	2026-04-27	2026-06-12	https://platform.openai.com/docs/api-reference/chat
All providers	2026-04-26	2026-06-12	—

S3 Coverage

Per-cell coverage is not uniform across the roster: free and rate-limited endpoints returned fewer admissible responses than the metered APIs, and one candidate (Gemini 2.5 Flash on early native batches) and a sixth judge (Command R+) hit provider quotas. We report realized coverage rather than silently dropping under-covered cells. Table S3 gives the model × task English-prompt matrix (column totals equal the by-task counts in the main text); Table S4 gives per-country English, native, and total counts with tier and pre-specification set; Table S5 lists the native-language pairs used for H2. The complete model × country × task coverage CSV, with the quota or exclusion reason for every shortfall, is included in the release.

Table S3. Coverage matrix: realized English-prompt confirmatory responses per model × task (T1 standard, T2 local datum, T3 health synthesis, T4 instruments, T5 recommendation). No cell is silently dropped; shortfalls are quota-driven (Methods).

Model	T1	T2	T3	T4	T5	Total
GPT-5	156	156	156	156	156	780
GPT-5-mini	156	156	156	153	156	777
Llama 3.1 8B	156	156	155	156	154	777
DeepSeek-V3	100	100	100	100	100	500
Command R+	100	100	100	100	100	500
Phi-4 14B	100	100	100	100	100	500
Qwen3 14B	100	100	100	100	100	500
Llama 4 Scout	100	100	100	100	100	500
Qwen3 32B	100	100	100	99	99	498
Cabra-Mistral 7B	99	97	100	100	100	496
Gemini 2.5 Flash	98	98	98	98	98	490
Claude Haiku 4.5	98	98	98	98	98	490

Model	T1	T2	T3	T4	T5	Total
Llama 3.3 70B	98	97	96	92	91	474
GPT-OSS 120B	68	72	60	42	56	298
Total	1529	1530	1519	1494	1508	7580

Table S4. Coverage by country: tier (GN/GS), pre-registration status (P = pre-registered 15; E = post-registration extension), and realized confirmatory responses (English, native-language, total).

Country	Tier	Set	English	Native	Total
Colombia	GS	E	260	251	511
Chile	GS	E	260	241	501
Portugal	GN	E	261	240	501
Angola	GS	E	260	240	500
Argentina	GS	P	355	139	494
India	GS	P	352	138	490
Mexico	GS	P	329	140	469
Peru	GS	P	316	141	457
Brazil	GS	P	260	141	401
Germany	GN	P	356	0	356
Indonesia	GS	P	356	0	356
Bangladesh	GS	P	355	0	355
Egypt	GS	P	355	0	355
Kenya	GS	P	348	0	348
Japan	GN	P	347	0	347
Nigeria	GS	P	317	0	317
Philippines	GS	P	316	0	316
United States	GN	P	312	0	312
South Africa	GS	P	302	0	302
United Kingdom	GN	E	265	0	265
Canada	GN	E	260	0	260
Australia	GN	E	260	0	260
France	GN	E	260	0	260
Italy	GN	E	260	0	260
South Korea	GN	E	258	0	258
Total (25)			7580	1671	9251

Table S5. Native-language confirmatory responses by language and country (H2 within-country English/native contrast).

Native language (Joshi)	Country	N (native)
Portuguese (4)	Portugal	240
	Angola	240
	Brazil	141
Spanish (5)	Colombia	251
	Chile	241
	Peru	141
	Mexico	140
	Argentina	139
Hindi (4)	India	138

S4 Country covariates

Table S6 reports the developmental and corpus-representation covariates used in the mechanism analysis (H4). Language-corpus measures (Joshi class; mC4/OSCAR sizes reported in the main text) index how much text exists *in* a country's language and are the candidate channel for the native-language penalty (H2). Country-coverage measures (English-Wikipedia article byte length, Wikidata sitelinks, Wikidata statements) index how broadly the encyclopedic corpus is *about* a country, independent of language, and are the candidate channel for the geographic gap (H1/H4). Because the geographic gap is measured *in English*, a residual North/South gap cannot be a language-corpus effect, which is why H4 isolates country coverage for H1 and language-corpus size for H2. All Wikidata entities are identified by a resolvable QID and were retrieved within the collection window.

Table S6. Country covariates. HDI: UNDP HDR 2023–24 (2022 data). Joshi: linguistic-resource class of the dominant official language (Joshi et al., 2020). Corpus measures from Wikimedia/Wikidata (entity QID resolvable): English-article byte length, Wikidata sitelinks (language editions), and Wikidata statements.

Country	Tier	HDI	Joshi	EN-wiki bytes	Sitelinks	WD stmts
Germany	GN	0.950	5	207,810	403	1402
Australia	GN	0.946	5	233,095	395	1674
United Kingdom	GN	0.940	5	368,888	391	1085
Canada	GN	0.935	5	281,016	393	1346
South Korea	GN	0.929	4	275,291	350	746
United States	GN	0.927	5	383,914	424	1710
Japan	GN	0.920	5	205,844	414	995
France	GN	0.910	5	279,498	413	1196
Italy	GN	0.906	4	305,234	405	1039
Portugal	GN	0.874	4	324,038	369	751

Country	Tier	HDI	Joshi	EN-wiki bytes	Sitelinks	WD stmts
Chile	GS	0.860	5	232,108	379	689
Argentina	GS	0.849	5	267,662	369	881
Mexico	GS	0.781	5	263,757	398	1286
Peru	GS	0.762	5	239,544	332	867
Brazil	GS	0.760	4	301,844	384	1151
Colombia	GS	0.758	5	303,401	363	774
Egypt	GS	0.728	5	284,547	387	757
South Africa	GS	0.717	5	270,861	365	750
Indonesia	GS	0.713	3	226,349	362	1005
Philippines	GS	0.710	5	493,981	331	963
Bangladesh	GS	0.670	3	275,786	343	814
India	GS	0.644	5	310,681	399	1679
Kenya	GS	0.601	5	216,189	334	699
Angola	GS	0.591	4	180,098	322	676
Nigeria	GS	0.548	5	266,041	349	1194

S5 Ground-truth registry (T1)

The validity of the geographic gradient (H1) requires that ground truth be equally available and equally verifiable across countries. For T1 (the binding annual $PM_{2.5}$ standard) we anchored every country to an official primary source by documentary audit: the source text was retrieved, the value was read verbatim, and the entry records the value, status, source-of-record, a resolvable URL, and the retrieval method (Table S7). Three countries have *no* national $PM_{2.5}$ standard, confirmed from the official text (Nigeria's NESREA regulates PM_{10} only; Argentina has no binding federal value; Angola has no national air-quality legislation); in those cases a faithful model must decline to state a non-existent threshold rather than fabricate one. This provenance is more reproducible than expert assertion: any reader can re-verify each value from the cited official source. No reference in the registry is unverified, and no value was carried over from secondary sources without confirmation against the primary text.

Table S7. Ground-truth registry for T1 (binding annual $PM_{2.5}$ standard), all 25 countries, anchored to official primary sources by documentary audit. All entries are of source type *official primary* unless noted. Status VERIFIED = value read from the official text; VERIFIED_NO_STANDARD = official confirmation that no national standard exists; PARTIALLY_VERIFIED = a revision is known to exist but its value could not be confirmed in an accessible official source. Full URLs and verbatim excerpts are in the released registry JSONL.

Country	Value	Status	Official source
Angola	NO national air-quality legislation / standard	VERIFIED_NO_STANDARD	No national ambient air quality regulation (World Bank/WHO note)

Country	Value	Status	Official source
Argentina	NO single binding federal $PM_{2.5}$ standard (Ley 20.284/1973 framework; $PM_{2.5}$ limits set subnationally)	VERIFIED_NO_STANDARD	Ley 20.284/1973
Australia	8 μg/m^3 (annual; 7 target 2025)	VERIFIED	National Environment Protection (Ambient Air Quality) Measure
Bangladesh	15 μg/m^3 (annual) per Gazette S.R.O. 220-Law/2005; a 2022 revision exists but the revised value is not confirmed in an accessible official source	PARTIALLY_VERIFIED	Bangladesh Gazette S.R.O. 220-Law/2005 (DoE)
Brazil	17 μg/m^3 (annual, PI-2)	VERIFIED	CONAMA 506/2024 Annex I, DOU 09/07/2024 Ed.130 p.133
Canada	8.8 μg/m^3 (annual, CAAQS 2020)	VERIFIED	Canadian Ambient Air Quality Standards (CCME)
Chile	20 μg/m^3 (annual)	VERIFIED	D.S. 12/2011 MMA Chile
Colombia	25 μg/m^3 (annual, in force since 2018; 15 from 2030; 24h=37)	VERIFIED	Resolucion 2254/2017 MinAmbiente Tabla 1 & 2 (verbatim official scan)
Egypt	$PM_{2.5}$ limit set under Law 4/1994 (amended 9/2009, 105/2015) Executive Regulations (official text in Arabic; value not extracted from an accessible official source)	PARTIALLY_VERIFIED	Environmental Law 4/1994 (EEAA)
France	25 μg/m^3 (annual, in force; 10 from 2030)	VERIFIED	EU Ambient Air Quality Directive 2008/50/EC; (EU) 2024/2881
Germany	25 μg/m^3 (annual, in force 2026; 10 from 2030)	VERIFIED	EU Ambient Air Quality Directive 2008/50/EC; revised (EU) 2024/2881
India	40 μg/m^3 (annual)	VERIFIED	CPCB National Ambient Air Quality Standards (2009)
Indonesia	15 μg/m^3 (annual)	VERIFIED	PP No. 22/2021 Lampiran VII (Indonesia)
Italy	25 μg/m^3 (annual, in force; 10 from 2030)	VERIFIED	EU Ambient Air Quality Directive 2008/50/EC; (EU) 2024/2881
Japan	15 μg/m^3 (annual)	VERIFIED	Ministry of the Environment Japan, Environmental Quality Standards (2009)
Kenya	35 μg/m^3 (annual; 24h 75)	VERIFIED	First Schedule, EMCA (Air Quality) Regulations 2014 (Kenya Gazette Vol. CXVI No.46)

Country	Value	Status	Official source
Mexico	10 $\mu g/m^3$ (annual)	VERIFIED	NOM-025-SSA1-2021, Diario Oficial de la Federacion 27/10/2021
Nigeria	NO national $PM_{2.5}$ standard (PM10 annual=60 $\mu g/m^3$, 24h=150)	VERIFIED_NO_STANDARD	NESREA National Environmental (Air Quality Control) Regulations, Schedule XIII (Ambient Air Quality Standards)
Peru	25 $\mu g/m^3$ (annual)	VERIFIED	D.S. 003-2017-MINAM (Peru)
Philippines	25 $\mu g/m^3$ (annual, 1-year guideline)	VERIFIED	DENR DAO (RA 8749 Clean Air Act); 25 ug/Nm3 annual / 50 ug/Nm3 24h
Portugal	25 $\mu g/m^3$ (annual, in force; 10 from 2030)	VERIFIED	EU Ambient Air Quality Directive 2008/50/EC; (EU) 2024/2881
South Africa	20 $\mu g/m^3$ (annual)	VERIFIED	SA NAAQS under NEMAQA (Act 39/2004), 2012
South Korea	15 $\mu g/m^3$ (annual)	VERIFIED	Ministry of Environment, Korea
United Kingdom	10 $\mu g/m^3$ (annual, target by 2040)	VERIFIED	Environmental Targets (Fine Particulate Matter) (England) Regulations 2023
United States	9.0 $\mu g/m^3$ (annual, primary NAAQS, 2024)	VERIFIED	EPA Final Reconsideration of PM NAAQS, Federal Register 2024-02637

S6 Composite outcome and scoring rubric

Every response was scored by the primary judge against the ground-truth registry entry (T1, T2, T4) or the validated rubric (T3, T5), producing five subcomponents each normalised to $[0, 1]$:

1. Factual accuracy against the registry value (binary for T1; partial credit for T2–T4).
2. Contextual completeness scored against a pre-validated rubric (T3, T4).
3. Citation quality, verified via DOI/official-source check.
4. Calibration, the alignment between stated confidence and correctness.
5. Absence of hallucination, coded binarily (0 hallucinated, 1 faithful).

The primary composite uses a pre-specified weighting (factual 0.30, contextual 0.25, citation 0.15, calibration 0.15, hallucination 0.15). As a sensitivity analysis the headline effects are recomputed under two alternative weightings — equal weights and a PCA-derived set (the first principal component of the five subcomponents) — and a conclusion is treated as robust only if its direction holds under all three; all headline effects are stable (tier gap +4.9 to +4.1–+5.1 pp; HDI gradient $\rho = 0.34$–0.37; factual floor δ from -0.44 to -0.47; regional-model deficit δ from -0.47 to -0.51). The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test. Each scored record also stores the judge's free-text rationale; for example, the `gpt-5-mini` judge scored a Llama 4 Scout reply on Brazil's T1 standard 0.15 with the rationale that it "gives wrong regulation (491/2018 vs correct 506/2024) and wrong current value… included a specific but incorrect citation and fabricated timing," illustrating how the factual-accuracy and hallucination subcomponents discriminate substantive failure.

S7 Judge-panel reliability

Reliability is computed on every item the panel scored — the same items the reported effects are computed from — rather than on a stratified calibration subset. All 3,190 responses requiring a judge were independently scored by all three panel members, chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement. The panel replaces the single-judge design of the original protocol, in which the primary judge (`gpt-5-mini`) was itself among the evaluated models.

Table S8. Inter-judge reliability on all 3,190 panel-scored items. The operative quantity is the reliability of the panel *mean*, $\mathrm{ICC}(2,3)$; by the Spearman–Brown relation it is high even when single judges agree only moderately. Source: `panel_reliability_full.json`.

Quantity	Value
Judges (k)	3 (Google, Anthropic, DeepSeek)
Items scored by all judges	3,190
Krippendorff's α (interval)	0.527
Single-judge $\mathrm{ICC}(2,1)$	0.558
Panel-mean $\mathrm{ICC}(2,3)$	0.791
Pairwise Pearson: Claude Sonnet | DeepSeek-V3	0.801
Gemini 2.5 Pro | Claude Sonnet	0.663
Gemini 2.5 Pro | DeepSeek-V3	0.654
Mean composite by judge: Gemini 2.5 Pro	0.617
Claude Sonnet 4.6	0.451
DeepSeek-V3	0.379

Pairwise agreement is strongest between Claude and DeepSeek and lower for the systematically more lenient Gemini. Because every reported effect is a between-group contrast (tier gap, language penalty, task floor), a judge's leniency offset cancels and does not threaten those contrasts — which is precisely why the analysis uses the panel *mean* rather than any single judge, whose reliability alone would be $\mathrm{ICC}(2,1) = 0.56$.

S8 Statistical methods and full confirmatory test outputs

The three primary confirmatory tests form a single family (F1) corrected together with the Bonferroni–Holm step-down procedure ($\alpha = 0.05$). Secondary confirmatory tests are reported unadjusted within family; declared exploratory tests use Benjamini–Hochberg FDR at $q = 0.10$. Rank correlations use Spearman ρ; monotonic-trend robustness uses the Mann–Kendall test; the persona effect uses a country-level permutation test (5,000 permutations). All outputs below are regenerated by `code/analysis/formal_tests.py`, `robust_tests.py`, and `h4_corpus_mechanism.py`.

Table S9. Primary family (Bonferroni–Holm): pre-specified 15-country values beside the post-registration 25-country extension. H1-by-HDI is the one primary test that rejects its null at $n = 25$; it remains below the pre-specified $\rho \geq 0.55$ effect-size threshold in both samples.

Test	15 countries (pre-specified)	25 countries (extension)
H1 Spearman ρ(acc, HDI)	+0.093, ns	+0.365 ($p = 0.072$), ns
H1 Mann–Kendall (by HDI)	ns	$p = 0.072$, ns
H1 Spearman ρ(acc, Joshi)	+0.028, ns	−0.061, ns (development, not linguistic)
H1 vs threshold $\rho \geq 0.55$	not met	not met
H4 partial ρ(acc, coverage | HDI)	ns	+0.177 ($p = 0.41$), ns
H6 DiD persona (GS−GN)	ns	+0.65 pp, perm $p = 0.269$, ns

Table S10. H4 corpus mechanism, multi-measure (25 countries). Country-coverage breadth (Wikidata sitelinks) predicts accuracy at the zero-order level and survives multiplicity correction across the three coverage proxies, but attenuates to non-significance after adjusting for development (HDI); we therefore report it as suggestive/exploratory, not as an established mechanism. The pre-specified language-edition proxy is null.

Measure	Family	ρ(acc, log measure)	partial | HDI
English-Wikipedia bytes	country	+0.153 ($p = 0.47$)	+0.132 ($p = 0.54$)
Wikidata sitelinks	country	+0.362 ($p = 0.075$)	+0.177 ($p = 0.41$)
Wikidata statements	country	+0.296 ($p = 0.15$)	+0.180 ($p = 0.40$)
Wikipedia-edition count	language	+0.142, ns	ns

S9 Additional robustness and corroboration analyses

The headline effects were re-tested under several pre-specified analyses, all regenerated by `code/analysis/weigh` `glmm_and_manipcheck.py`, `bayesian_reestimation.py`, and `mediation_h4.py` (wired into `run_all.py --conf`

Composite-weighting sensitivity. Recomputing every effect under the primary author-specified weighting, equal weights, and a PCA-derived weighting (first principal component of the five sub-components) leaves all four headline conclusions stable in direction (Table S12).

Mixed model and Bayesian re-estimation. A Gaussian linear mixed model (`statsmodels` MixedLM, REML) with crossed random intercepts for country and model gives a Global South effect of $\beta = -0.066$ (SE $= 0.036$, $p = 0.069$), marginal rather than conventionally significant because it charges the large between-country heterogeneity to the error term. A Bayesian hierarchical re-estimation with weakly informative priors (`pymc`; crossed country/model intercepts, `random_seed=42`) agrees on magnitude and is decisive about the sign: posterior mean −0.052, 94% HDI $[-0.092, -0.016]$, $P(\beta < 0) = 0.994$ ($\hat{R} = 1.001$).

E-values. The tier gap carries an E-value of 1.70 at the point estimate and 1.31 at the confidence limit nearest the null (VanderWeele & Ding 2017 approximation from the standardized effect); the second is the operative number, and it is a low bar. No E-value is reported for the development gradient: its interval already includes the null, so the E-value at that limit is 1.00 and no confounder at all is required.

Table S11. Robust secondary findings (25 countries), with leave-one-country-out (LOCO) and range-restriction checks.

Finding	Statistic
Tier gap (GN−GS)	+5.1 pp, 95% CI $[+1.6, +8.5]$; permutation $p = 0.020$
Tier gap, LOCO range	$[+4.5, +6.0]$ pp, stable
H1 leave-one-country-out range	$\rho \in [+0.31, +0.49]$: never reaches the pre-specified 0.55
Task floor (T1+T2 vs T3–T5)	0.417 vs 0.623, Cliff's $\delta = -0.47$
H3 Cabra-Mistral vs rest	0.347 vs 0.549, Cliff's $\delta = -0.47$
H2 native vs English (all)	$\Delta = -4.8$ pp, Wilcoxon $p = 3 \times 10^{-15}$ ($n = 839$)
Spanish / Portuguese / Hindi	−4.4 / −3.9 / −11.1 pp, all significant
H5 open vs closed-accessible	+13.3 pp closed advantage (descriptive)

Table S12. Headline effects under three composite weightings (English confirmatory, n = 7,580). Direction is stable throughout.

Weighting	Tier gap (pp)	ρ(HDI)	Floor δ	H3 δ
Author-specified (primary)	+5.06	+0.365	−0.471	−0.471
Equal	+4.85	+0.530	−0.588	−0.524
PCA-derived	+4.84	+0.375	−0.494	−0.473

Exploratory mediation (H4). A standardized country-level path model (`semopy`, $n = 25$): HDI $\rightarrow$ sitelinks $a = +0.67$; sitelinks $\rightarrow$ accuracy $b = +0.27$; direct $c' = +0.31$; indirect $ab = +0.18$ with bootstrap 95% CI $[-0.27, +0.59]$ (includes zero). Country coverage is thus a suggestive but not established mediator, consistent with the main-text framing.

Persona manipulation check. Across 4,349 persona-condition responses, 50.2% explicitly acknowledge the public-manager role frame, so the null persona effect (H6) is not an artifact of the frame being universally ignored.

S9.1 Tier gap under alternative country partitions

The Global North/South split follows the UNCTAD classification. A gap visible under one taxonomy alone would be an artefact of that taxonomy, so we re-estimated it under three development-based partitions of the same 25 countries (Table S13). The gap is significant under UNCTAD and non-significant under every partition drawn on HDI, which is consistent with the null HDI gradient reported in the main text.

Table S13. Tier gap under alternative partitions of the same 25 countries. Permutation test with country as the unit, 10,000 relabellings.

Partition	Gap (pp)	GN/GS	p
UNCTAD North/South (used throughout)	+5.06	10/15	0.020
HDI below the sample median (0.781)	+2.39	13/12	0.269
HDI $<$ 0.800 (UNDP "very high" cutoff)	+2.92	12/13	0.182
HDI $<$ 0.700	+1.53	20/5	0.575

S10 Deviations from the pre-specified analysis plan

The analysis plan required that departures be logged and classified, and set a ceiling on them: confirmatory status is retained only if deviations affect no more than 15% of planned cells. The table below reports every departure we identified. The delivered scope is 27.5% of the planned response count, so the ceiling is exceeded by a wide margin, and the study is reported as exploratory throughout, as described in the Study protocol statement.

Table S14. Departures from the pre-specified plan, with the effect of each on interpretation. "Planned" quotes the analysis plan; "Delivered" is what the released artefacts contain.

Item	Planned	Delivered	Effect on interpretation
Response count	33,600 responses (600 prompts $\times$ 14 models $\times$ 2 personas $\times$ 2 replicates)	9,251 scored responses (27.5%)	Triggers the plan's own 15% ceiling. The study is reported as exploratory.
Public registration	Plan deposited in a public registry before confirmatory collection	Plan fixed in a version-controlled document; never deposited	Timing cannot be verified independently of repository history. We do not claim pre-registration.
Rubric pilot	50-prompt pilot scored by two independent human raters, Krippendorff's $\alpha \geq 0.70$ required before scaling	Not run. No rater-level α exists	Rubric reliability is evidenced only after the fact, on confirmatory data, at $\alpha = 0.527$ across the 3,190 items scored by all three panel judges, below the threshold the plan had set.
Judging	Three-judge LLM ensemble plus external human annotators blind to hypotheses	Three-vendor panel on every item that required a judge (3,190 items); code adjudication where the answer is a checkable register value; no human layer	The ensemble is as planned, but no human gold standard exists anywhere in the study. Reliability rests on model-to-model agreement, $\mathrm{ICC}(2,3) = 0.79$.
Ground-truth registry	Per (country, task) entry with source class, resolvable URL, gold value, reference date, human validator and validation flag	Registry covers T1 for all 25 countries; T2, T3 and T4 were built in a later revision from single international sources (WHO AAQD v6.1, WHO GHO AIR_41, UNEP GAAPL Appendix 1); no human validator field exists	Ground-truth provenance is now anchored for four of five tasks but is not uniform in source class, and the claim of a per-(country, task) registry did not hold in the original submission draft.

Item	Planned	Delivered	Effect on interpretation
Country sample	15 countries, fixed in advance	15 extended to 25 after the pre-specified analysis was run	The extension was made to gain power for the gradient, which nonetheless does not reach significance at $n = 25$ under the corrected scoring ($\rho = 0.37$, $p = 0.080$). Every effect is reported alongside its 15-country value.
Scoring instrument	Single primary judge (GPT-5-mini) scoring all tasks against the registry	T2 and T3 adjudicated by code against official registers where the returned value is extractable and plausible (62.5% and 83.6% of cells); the residual and all of T4 re-scored by the three-vendor panel mean; T1 and T5 unchanged	Changes the substantive conclusions on the same responses. The development gradient falls from $\rho = 0.51$ ($p = 0.004$) to $\rho = 0.37$ ($p = 0.073$); the native-language penalty grows from -2.1 pp to -4.8 pp and becomes significant in all three languages. The findings are reported in the revised order.
Ordering of findings	Plan treated H1 (geographic gradient) as the primary hypothesis	H2 (native-language penalty) is reported as the principal finding and H1's gradient as exploratory	Post-hoc reordering, driven by the corrected scoring rather than by the data being re-examined for a better story. Both hypotheses were pre-specified and both are reported; only the prominence changed.
Judge prompt defect	—	An early T2 judge prompt asked the judge to compare a value against a tolerance band expressed in words, requiring it to do the arithmetic. Concordance on those items was $r = -0.04$; pre-computing the range raised it to $r = +1.00$	The 274 scores collected under the ambiguous wording were discarded and recollected. Reported because the defect first presented as genuine inter-judge disagreement.

Item	Planned	Delivered	Effect on interpretation
Corroborating analyses	Mixed model, Bayesian re-estimation and E-values reported alongside the primary tests	Recomputed on the corrected scoring	All three are weaker than the single-judge versions. The mixed model falls from $p = 0.007$ to $p = 0.069$; the Bayesian posterior from $P(\beta < 0) = 1.00$ to 0.994; the tier-gap E-value from 1.85 to 1.70 at the point and 1.31 at the confidence limit. The tier gap is reported as supported but specification-sensitive.
E-value reporting	E-value of the point estimate for tier gap and HDI gradient	E-value of the point estimate *and* of the confidence limit nearest the null, for the tier gap only	Corrects a defect of our own. Reporting only the point estimate overstates robustness. The gradient's E-value is withdrawn: the association is not significant, so its nearest-null E-value is 1.00 and no confounder is required to explain it away.
H4 adjustment set	Partial Spearman adjusting for HDI *and* log GDP per capita	Originally adjusted for HDI only; GDP did not appear in the analysis code. Now executed as specified	Resolved. The specified adjustment gives $\rho = +0.02$ against $+0.04$ published; the two control sets agree because HDI and log GDP correlate at $\rho = 0.97$ across these countries.
Native-language rendering	Human native-speaker translation with independent back-translation and bilingual adjudication	LLM translation, audited after the fact by a back-translation census of all 90 native prompts: a different model back-translates, two further models audit against the English original on five items, one flag marks the prompt divergent	Partially resolved. No human translator and no bilingual adjudication, so the plan is not met. But the artefact the plan guarded against is measured and absent: 90/90 preserve the country and the requested quantity, 88/90 the question, and H2 is unchanged among verified-faithful prompts (-4.6 pp; fidelity does not predict effect size, permutation $p = 0.42$).
Extra H2 variants	Two additional question variants per (country, task), rendered in both languages	Generated but never collected or judged	H2 rests on the principal pairs only; the planned cell count for H2 was not reached.

Item	Planned	Delivered	Effect on interpretation
Model coverage	All 14 stacks across all countries and tasks	Coverage is not uniform: two stacks are absent from several countries after provider rate limiting and credit exhaustion	Per-country means are computed over different model sets. A balanced-subset check is reported in Section S9.

A further property of the response corpus, not a departure from the plan but a consequence of how collection was retried, belongs here because it bears on reproducibility. Of 9,760 unique (model, prompt, replicate) keys, 2,147 (22.0%) have more than one stored response, produced by re-runs on different dates; restricting to records the scoring pipeline admits gives 9,140 keys and 2,095 duplicates (22.9%). Earlier drafts reported 4,976 and 1,003 here, figures that predate the extension to 25 countries and that we could not reproduce from the released artefacts; the counts above are regenerated by `code/analysis/robustness_extra.py`. The stored answers differ substantially between runs; in one case the same prompt and model yielded 307 and 3,180 characters. The scoring pipeline consumes the first record encountered in alphabetical file order, so which of several stored answers enters the analysis is determined by filename ordering rather than by an analytical rule. We report this rather than leave it implicit.

S11 Prompt construction, persona template, and verbatim examples

S11.1 Representative failure modes, verbatim

Table S15. Box 1 — Representative failure modes (verbatim, lightly truncated).

Failure mode	Example
Fabricating a standard that does not exist	Angola has no national norm. Gemini 2.5 Flash asserted an annual standard of "25 μg/m^3" in one replicate and "15 μg/m^3" in another. GPT-5 asserted "15 μg/m^3" for Nigeria, which regulates PM_{10} only (NESREA). A faithful response would decline to state a value that does not exist.
Missing the binding value (the T1 floor)	Brazil's official annual standard is 17 μg/m^3 (CONAMA 506/2024). The modal answer was 15 μg/m^3 (42 responses), with 10, 20 and 25 also common; the official value was almost never returned.
Degradation in the native language (H2)	India's annual standard is 40 μg/m^3. GPT-OSS 120B returned the correct value in English and an *empty* answer to the same question in Hindi.

Prompts were authored in English for the air pollution policy domain with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set. The plan required a two-rater pilot to reach Krippendorff's $\alpha \geq 0.70$ before scaling; that pilot was not run, and the deviation is logged in Section S10. Native-language renderings (Portuguese, Spanish, Hindi) were produced by a high-quality LLM translator (Claude Sonnet 4.6) instructed to preserve the meaning, the persona frame, and proper nouns/acronyms, with the factual ground truth held in English (language-invariant). Human native-speaker translation and back-translation verification were not performed, and this is noted as a limitation in the main text. Every factual claim in a ground-truth entry was required to resolve to an official-source URL or a peer-reviewed DOI before the corresponding prompt was admitted (the project Citation Verification Protocol applied at the ground-truth generation stage, not only at the manuscript stage).

Persona manipulation. Persona is a within-prompt factor with two levels that hold the substantive question identical. The `neutral` level presents a plain information request; the `public_manager_env` level prepends a fixed role frame. The contrast is shown verbatim for Brazil, T1:

> *Neutral.* "What is the current national annual mean ambient air quality standard for PM2.5 in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences."
>
> *Public environmental manager.* "*I am the municipal environmental management official of São Paulo, Brazil, and I need to brief decision-makers.* What is the current national annual mean ambient air quality standard for PM2.5 in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences."

The corresponding registry ground truth is: "Brazil's national $PM_{2.5}$ standards are set by CONAMA Resolution 506/2024 (which superseded the relevant articles of Resolution 491/2018). The stage in force in 2026 is Intermediate Target PI-2; the Final Target adopts the WHO 2021 annual guideline of 5 μg/m^3" (source: CONAMA Resolution 506/2024). Native-language prompts apply the same template, translated; for example the Hindi T1 prompt for India is the faithful Devanagari rendering of the neutral English item, reproduced verbatim in the released prompt set. The complete prompt set (370 confirmatory items: 25 countries $\times$ 5 tasks $\times$ persona, plus native-language pairs) is released as `prompts_confirmatory.jsonl` with the per-item ground truth, source, and rubric.

References

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